SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

Duration : 45 Minutes

QUESTIONS: 2 **Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
   Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

INTELLIGENT AGENTS IN A SMART HOME AUTOMATION SYSTEM

INTRODUCTION:

An intelligent agent is a system that observes its environment through sensors and Performs actions using actuators to acheive specific objectives. In a smart home, intelligent agents control lighting, temperature and security based on motion detection, environmental conditions, time and user preferences

Types of Intelligent Agents

1 SIMPLE REFLEX AGENT

2 MODEL - BASED AGENT

3 GOAL - BASED AGENT

4 UTILITY - BASED AGENT

1. Simple Reflex Agent

A Simple reflex agent Select actions based only on the current input using Predefined condition - action rules.

- If motion is detected, the light is switched ON.

- Simple reflex agent: Reacts to current sensor inputs, such as turning when a wall is detected. without memory, it may revisit locations.
- Model-based agent: Remebers explored cells and obstacles. allowing the robot to update its internal maze-map.
- Goal-based agent: plans action specifically to reach the exit.
- Utility-based agent compares routes based on travel cost, battery usage, time and risk

CONCLUSION:

UCS is suitable for finding the least cost path, whereas IDS is useful when memory is limited and movements have equal costs. combining an appropriate search strategy with model-based and goal-based reasoning allows the robot to explore an unknown maze systematically, avoid obstacles and reach the exit efficiently

Working Steps:

1. Start with a depth limit of zero
2. Search from the initial position with in the Current limit
3. If the goal is not found, increase the limit by one
4. Repeat the search
5. Stop when the goal is found or the search space is exhausted

For example, if the exist is four steps away, IDS searches up to depth 0, then 1, 2, 3, and finally 4.

Advantages:

- Requires less memory that breadth-first Search
- Find the shallowest solution when all actions have equal costs
- useful for large spaces with limited memory.

3 Relationship to Intelligent Agents

Different intelligent agents affect how the robot make decisions.

the goal is reached.

The path cost is calculated as

$$g(n) = g(\text{parent}(n)) + c$$

Here, $g(n)$ is the total cost of reaching node n and c is the cost of the latest movement.

Advantages:

- Find the least-cost path when movement costs are non-negative and Standardness conditions hold
- Handle different movement costs
- Suitable for avoiding expensive or risky routes when these are represented by appropriate costs.

2 Iterative Deepening Search (IDS)

IDS repeatedly performs depth-limited depth-first search, increasing the depth limit after every unsuccessful attempt

SOLVING AN UNKNOWN MAZE USING UCS AND IDS

INTRODUCTION:

A robot placed in an unknown maze must find a path from its starting position to the exit without knowing the complete layout. It uses sensors to detect obstacles and explores possible routes using seach algorithms

uniform cost search (UCS)

UCS expands the node with the lowest accumulated Path cost. It is suitable when different movements have different cost

1. Place the starting position in a priority queue with cost zero
2. Select the position with the lowest path cost
3. Check whether it is the goal
4. Identify the valid neighboring positions
5. calculate the cost of reaching each neighbor
6. update the priority queue and Prepare until

Advantage: Balance multiple objectives, impr
energy efficiency and supports personalized deci
Limitation: Designing a suitable utility functi
is difficult and incorrect priorities may produce
Poor decisions

Most effective Approach

- Simple reflex rules provide immediate response to motion and Security events
- Model-based reasoning remembers room Occupancy and environmental conditions
- Goal-based Planning walk. Toward comf

energy saving and security

Conclusion:

A hybrid intelligent agent provides the best solution because it combines fast respons environment awareness, Planning and utility optimization.

Goal - Based Agent

A goal - based agent selects actions that help achieve a specific objective. It goals may include maintaining a comfortable temperature, reducing electricity consumption and securing the house when nobody is present.

For example, before the user arrives home, the agent can turn on the lights and adjust the temperature

Advantages: Support planning and flexible decision making

Limitations: Planning may require more computation, and conflicting goals can be difficult to manage.

Utility - Based Agent

A utility - based agent evaluates possible actions and select the one that provides the highest overall benefit. It consider comfort, energy consumption, security and user preferences For example, when the user leaves home, it may switch off unnecessary light, reduce air-conditioning power and activate the security system.

- If the temperature exceeds a specific limit, the air conditioner is activated.
- If an intruder is detected, an alarm is triggered.

ADVANTAGES: Fast response, simple design, and low computational requirements.

LIMITATIONS: Cannot remember previous events or learn user preferences. It may make unsuitable decisions in complex situations

2 MODEL- BASED AGENT

A model- base agent maintains an internal State of the environment using current and Previous observations. It can remember room occupancy, temperature changes and the status of doors and windows.

For example it can wait briefly before Switching off lights when a person leaves a room. It can also check whether a window is open before activating the air conditioner

Advantages: uses past information and handles Partially observable enviroments

Limitations Requie memory and can accurate internal model.